Wood Provision Ecosystem Service Accounts

European Union, 2000 to 2024 series update: processing methods and results

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Summary

We provide estimates of the wood provision ecosystem service for the 27 Member States of the European Union in 2000, 2006, 2012, 2018 and 2024, following the accounting method of INCA Part III (La Notte et al., 2021), which updates INCA Part II (Vallecillo et al., 2019). The actual flow is the net annual increment of the forest available for wood supply, which falls from 955 million cubic metres in 2000 to 749 million cubic metres in 2024. Part of that fall comes from a change of source within the series, as Section 2.2.2 shows. Removals stay below the increment in every Member State in the reporting years up to 2018, so the sustainable flow equals the removals in those years. In 2024, 3 Member States show overuse, adding up to 6.2 million cubic metres, and those flags rest on values carried forward from earlier years. At constant 2010 prices, the actual flow is worth 25.6 billion euro in 2000 and 17.2 billion euro in 2024, and the overuse of 2024 is worth 146 million euro. All quantities are reported by NUTS0, NUTS2 and NUTS3 region and mapped at 100 m.

1 Introduction

Wood provision is the ecological contribution to the production of timber that can be harvested and used as a raw material. Its underlying process is the natural growth of a biotic resource, so the accounts measure it through the net annual increment (NAI) of standing timber in the forest available for wood supply (La Notte et al., 2021). NAI is the average annual growth in volume of living trees less the average annual losses from natural mortality, measured here in cubic metres over bark (Eurostat, 2026a). Forest available for wood supply is forest without environmental, social or economic restrictions with a significant impact on the supply of wood.

Services with a regeneration rate can be used faster than they renew, so the accounts also compare use with regeneration. Overuse is the part of the use that exceeds regeneration, and the sustainable flow is the part that stays within it.

This report describes how we estimated those quantities and valued them, why we took each decision along the way and what the estimates show. The calculations are carried out by a series of scripts in the folder https://github.com/renatovargas/jrc-wood-provision/tree/main/scripts, listed in Table 1 in the order in which they run. Each step of the Methods section points to the entry of its script in Annex I, which explains what the script does and links to its code. The file run_all.R runs the whole series or any of its stages, and any single script can run

again when only one input changes. The tables and the 100 m maps can be retrieved from the Downloads page of this site.

Table 1: Scripts of the pipeline, in the order in which they run. Each name links to its entry in Annex I.

Script	Purpose
00_prepare_gdmp_annual.R	Annual productivity from the ten-day files, run by hand
01_prepare_nuts.R	Country boundaries
02_download_eurostat.R	Forest volumes, value of forest trees, roundwood exports and prices
03_prepare_carbon_budget.R	Increment estimated by the carbon budget model
04_extract_manual_annex.R	Published tables by Member State, for comparison
05_prepare_weights.R	Forest times productivity, at 100 m and 1 km
06_prepare_nuts_weights.R	Share of each region in the weight of its country
07_build_physical_accounts.R	Actual flow, sustainable flow and overuse by country
08_build_monetary_accounts.R	Unit values, value by country and the results workbook
09_write_maps_1km.R	Maps at 1 km, their checks and the map figures
10_write_maps_100m.R	Maps at 100 m
11_check_maps_100m.R	Comparison of the 100 m maps with the tables
12_compare_dmp_gdmp.R	Test of Dry Matter Productivity against Gross Dry Matter Productivity, run by hand

1.1 Sources of orientation

The method comes from Chapter 4 of INCA Part II and Section 6.2 of INCA Part III, and these reports settle any question they address. Section 6.2 of Part III replaces two steps of Part II. It reports the whole NAI as the flow, because the forest asset accounts already record natural growth as an addition to the stock, and because forest management does not change stand volume growth much within a range of thinning grades (La Notte et al., 2021). The ecosystem contribution coefficient of Part II is therefore dropped. It also replaces the valuation through the output of forestry with a valuation through export prices.

The INCA tool applies the same method and adds details that the reports do not state, such as the list of outliers described in Section 2.4.1. We took those details from it. The reports give no method for overuse of wood or for the sustainable flow, so those steps are decisions of this work, explained below.

2 Methods

2.1 Territory and reporting regions

The accounts cover the EU27 with the NUTS 2024 boundaries, at the country level (27 regions), NUTS2 (239 regions) and NUTS3 (1160 regions) (Eurostat GISCO, 2024) (see 01_prepare_nuts.R in Annex I). Malta has no forest available for wood supply and no forestry activity (Vallecillo et al., 2019), so its values are zero throughout.

2.2 Actual flow

The source of the actual flow is the table `for_vol_efa` of Eurostat, which collects the European Forest Accounts in thousand cubic metres over bark (Eurostat, 2026e) (see 02_download_eurostat.R in Annex I). Table 2 shows why it needs completing, since the NAI of forest available for wood supply is reported by few Member States before 2018.

Table 2: Member States out of 27 that report each series in `for_vol_efa`. FAWS is forest available for wood supply and FOR is total forest.

Series	Forest	2000	2006	2012	2018	2024
Net annual increment	FOR	26	26	26	26	2
Removals	FAWS	26	26	26	26	2
Removals	FOR	26	26	26	26	2
Net annual increment	FAWS	0	2	6	14	2

2.2.1 Filling the gaps

INCA Part III fills the gaps with the estimates of the carbon budget model (La Notte et al., 2021; Pilli & Grassi, 2021). These estimates are published in a workbook that combines the model with the forest accounts and gives the total NAI of each country from 2000 to 2019, with the values after 2016 held at their level of 2015 (see 03_prepare_carbon_budget.R in Annex I). The series refers to total forest, and the workbook also gives the share of available forest in the growing stock, so two versions of the fill are possible. The INCA tool uses a third, the NAI of total forest reported to Eurostat. Table 3 compares all three with the values by Member State printed in the annexes of Part III, which are the published reference for the method (see

04_extract_manual_annex.R in Annex I for the annexes and 07_build_physical_accounts.R in Annex I for the comparison).

Table 3: Rules to fill the gaps in the NAI of available forest, against the values of Annexes AA37, AA41 and AA45 of INCA Part III. The differences add up the absolute gap of each Member State, without the United Kingdom.

Rule	Year	EU27 (million m3)	Published (million m3)	Sum of differences (million m3)	Median difference (%)
Total forest reported to Eurostat	2000	912	923	112	13.2
Total forest reported to Eurostat	2006	900	866	83	7.1
Total forest reported to Eurostat	2012	855	850	81	9.4
Carbon budget model, total forest	2000	955	923	39	4.0
Carbon budget model, total forest	2006	916	866	64	7.4
Carbon budget model, total forest	2012	887	850	60	7.0
Carbon budget model, scaled to available forest	2000	855	923	69	5.7
Carbon budget model, scaled to available forest	2006	824	866	68	8.5
Carbon budget model, scaled to available forest	2012	800	850	64	7.7

The total forest fallback has the closest EU totals, but only because errors offset, as its country differences are the largest, at 112, 83, 81 million cubic metres in the three years. The total forest estimate of the carbon budget model gives 39, 64, 60 million cubic metres. It follows the method that Part III states and reproduces its member state values best, so a decision was made to use it. The NAI is therefore taken in this order. A value reported for available forest comes first. The carbon budget estimate comes next. Any remaining gap takes the mean of the nearest earlier and later values of the same country, and gaps at the end of the series repeat the last value. Table 4 shows what each reporting year rests on.

Table 4: Origin of the NAI of the 27 Member States in the reporting years. Malta is the entry with no supply.

Year	Reported	Carbon budget model	Carried	No supply
2000	0	26	0	1
2006	2	24	0	1
2012	6	20	0	1
2018	14	12	0	1
2024	2	0	24	1

The carbon budget estimate covers total forest and fills a statistic of available forest, so it probably overstates the flow in the countries that take it. Table 5 shows the size of this effect against the EU27 aggregates that Eurostat publishes, which it estimates itself. The accounts sit above the aggregate for available forest by 9 percent in 2012.

Table 5: EU27 NAI in the accounts, the Eurostat aggregates and the sum of the published Member State values without the United Kingdom, in million cubic metres.

Year	Accounts	Eurostat, available forest	Eurostat, total forest	INCA Part III
2000	955	891	912	923
2006	916	838	900	866
2012	887	814	871	850
2018	822	784	849	
2024	749			

Eurostat holds very few values for 2024. Table 6 gives the last year with an original value for each Member State, so the column of 2024 mostly repeats earlier years, and the method says

nothing about how far a value may be carried. A decision was made to carry the last value, since the increment changes slowly and any trend applied to it would add an assumption that the data cannot support.

Table 6: Last year with a value reported or estimated for the NAI, as a count of Member States.

Last year	Member States
2019	3
2023	21
2024	2

2.2.2 Changes of source within the series

A Member State that starts reporting available forest moves from the estimate of the model to its own figure, and the two can differ widely. Table 7 lists the Member States where this happens between two reporting years. France falls by 30 percent between 2006 and 2012 and Sweden by 29 percent between 2012 and 2018, and in both cases the model estimate of the later year is far from the reported figure. Part III has the same feature, since its value for France goes from 156 million cubic metres in 2000 to 93 million in 2012. We kept the series as the method builds it and did not adjust the model estimates to the reported levels, so changes over time within these countries should be read together with this table.

Table 7: Member States whose NAI changes from the model estimate to a reported value between two reporting years, in thousand cubic metres.

Country	From	To	NAI, from	NAI, to	Model, to	Change (%)
CY	2012	2018	254	55	253	-78
FR	2006	2012	132,498	93,230	127,389	-30
SE	2012	2018	130,469	92,708	133,291	-29
SK	2012	2018	16,252	12,764	18,532	-21
SI	2006	2012	8,440	6,950	4,253	-18
AT	2000	2006	36,936	30,820	38,596	-17
DE	2012	2018	111,379	93,747	123,591	-16
HR	2012	2018	10,202	8,893	10,696	-13
RO	2006	2012	31,908	35,818	58,171	12

Country	From	To	NAI, from	NAI, to	Model, to	Change (%)
IE	2012	2018	8,703	7,900	9,333	-9
LU	2006	2012	717	760	689	6
DK	2012	2018	7,747	8,152	7,122	5
BG	2000	2006	15,683	15,409	19,401	-2
PL	2012	2018	79,820	79,236	76,997	-1

2.3 Sustainable flow and overuse

The accounting reports class wood among the services with a regeneration rate that use can exceed (La Notte et al., 2021), and the handbook of ecosystem accounts describes overharvesting as removals above the NAI (Eurostat, 2026a). Neither gives a formula for wood, so we adapted the one of Part III for water purification, in which overuse is the difference between the actual and the sustainable flow and the threshold defines the sustainable flow. Regeneration is the threshold for wood, which gives these two equations for each country and year.

$$\text{Overuse} = \max(\text{Removals} - \text{NAI},\ 0) \tag{1}$$

$$\text{Sustainable flow} = \min(\text{Removals},\ \text{NAI}) \tag{2}$$

Removals are the removals of available forest in `for_vol_efa`, which Member States report more widely than the NAI, as Table 2 shows. They take the same rule for gaps, with reported removals of total forest in place of the model estimate. The two quantities add up to the removals in every country and year (see 07_build_physical_accounts.R in Annex I). A single year above the threshold can be a temporary peak, for example after a storm, so the table of results also gives the ratio of removals to NAI, and the handbook treats overharvesting as a matter of an extended period.

2.4 Monetary valuation

The value is the physical quantity times a unit value in euro per cubic metre. INCA Part III derives the unit value in three steps (La Notte et al., 2021). The output of forest trees in `for_sup_cp` (Eurostat, 2026d) divided by the NAI gives the unit value of the NAI, which few Member States report in the early years, as Table 8 shows. The roundwood exports of `for_basic` (Eurostat, 2026c), in value and in volume, give a unit value for most countries and years, but this price includes the harvesting and transport added on the way to the market. The ratio between the two then carries the export price back to the standing tree.

$$\text{Ratio}_{c,t} = \frac{\text{Unit value of exports}_{c,t}}{\text{Unit value of NAI}_{c,t}} \tag{3}$$

$$\text{Completed unit value}_{c,t} = \frac{\text{Unit value of exports}_{c,t}}{\widehat{\text{Ratio}}_{c,t}} \tag{4}$$

Where the ratio is missing, $\widehat{\text{Ratio}}$ takes the mean ratio of the country, or the EU mean if the country has no ratio at all (see 08_build_monetary_accounts.R in Annex I). Gaps in the export price take the same means. Means run over 2000 to 2023 for a country. The EU mean takes the mean of the countries in each year and then the mean of those years from 2012 to 2023, where coverage is better, as in the INCA tool.

Table 8: Member States out of 27 with a reported timber value and an export unit value.

Year	Timber value	Export unit value
2000	2	24
2006	3	23
2012	16	23
2018	17	23
2024	2	20

2.4.1 Screening the ratio

The ratio is very sensitive to small denominators. Eurostat records the output of forest trees in Croatia as 0.00 or 0.01 million euro, which yields a ratio near 90,746. With no screening, the EU mean of the ratio is 921, against a median of 3.3 among the ratios kept after screening, so every Member State that takes the EU mean ends up with a unit value close to zero.

The INCA tool removes all the timber values of Croatia, Finland, Greece, Portugal and Sweden and the export points of Denmark in 2014 and Greece in 2006 (European Union, 2026), and we did the same. The data used here are more recent than that list, so it does not catch everything. Lithuania now has a median ratio of 1,547. Belgium has a median of 0.89, which would make the standing tree worth more than the harvested roundwood, and Part III reports Belgium at a far lower value. A decision was made to extend the rule in a way that the data support. A Member State whose median ratio falls below 1 or above 100 loses all its timber values, and any single observation outside that range is dropped. The lower bound follows from the meaning of the ratio. Among the Member States left after the list of the INCA tool,

no ratio lies between 13 and 1,000, so any upper bound in that range gives the same result. Table 9 shows the outcome. After screening, the EU mean of the ratio is 3.52.

Table 9: Ratio of export to timber unit values by Member State, after the removal of export errors and before the screening of the ratio.

Country	Observations	Median	Minimum	Maximum	Treatment
HR	2	90,745.5	83,511.3	97,979.7	All timber values removed, as in the INCA tool
LT	4	1,547.3	1,182.8	1,617.9	All timber values removed
EL	2	67.6	67.3	67.8	All timber values removed, as in the INCA tool
PT	24	11.8	3.2	39.6	All timber values removed, as in the INCA tool
FI	8	11.0	2.9	18.6	All timber values removed, as in the INCA tool
ES	5	6.8	5.4	7.2	Kept
SE	13	5.6	4.3	10.0	All timber values removed, as in the INCA tool
LU	10	5.2	3.6	12.6	Kept
IT	12	4.8	2.6	11.4	Kept
RO	14	4.6	3.4	7.2	Kept
AT	25	4.6	3.0	7.4	Kept
LV	4	4.4	3.0	6.2	Kept
HU	2	3.5	3.3	3.7	Kept
SK	12	3.1	2.4	5.8	Kept
FR	13	2.7	2.1	4.1	Kept
CZ	10	2.7	2.0	2.9	Kept

Country	Observations	Median	Minimum	Maximum	Treatment
DE	12	2.7	2.2	5.1	Kept
PL	11	2.4	2.0	3.0	Kept
SI	13	2.2	1.8	2.5	Kept
NL	3	2.1	1.7	4.7	Kept
EE	2	2.0	1.8	2.1	Kept
IE	2	1.8	1.6	2.1	Kept
BG	11	1.3	0.7	1.8	Kept
BE	2	0.9	0.4	1.4	All timber values removed

Export prices have their own errors. A decision was made to drop a point that lies more than three times above or below the median of its country, a rule that also catches the Greek point of the INCA tool, while the Danish point has no data in this release. It removes 19 points, listed in Table 10, and 5 of them fall in reporting years. Among those, Croatia in 2012 stands at 10.4 euro per cubic metre against a median of 67.3, and Romania in 2024 at 286.3 euro against a median of 92.8. Table 11 shows how much of the result rests on each kind of ratio after these steps.

Table 10: Export unit values removed, in euro per cubic metre.

Country	Year	Unit value	Median of the country
BE	2014	465.6	86.2
BE	2015	648.6	86.2
BE	2016	448.5	86.2
BE	2021	25.6	86.2
BE	2023	13.6	86.2
BE	2024	27.0	86.2
EL	2001	107.1	31.6
EL	2006	0.9	31.6
EL	2007	1.0	31.6
EL	2008	1.9	31.6

Country	Year	Unit value	Median of the country
EL	2009	3.1	31.6
EL	2010	2.4	31.6
EL	2011	2.5	31.6
HR	2012	10.4	67.3
IE	2000	215.2	62.6
LT	2013	233.6	63.6
LU	2010	249.1	72.5
LU	2016	0.1	72.5
RO	2024	286.3	92.8

Table 11: Origin of the ratio of the 27 Member States in the reporting years.

Year	Observed	Country mean	EU mean
2000	1	16	10
2006	2	15	10
2012	9	8	10
2018	11	6	10
2024	2	15	10

2.4.2 Constant prices

Part III moves its other accounts to constant prices, so that the accounts follow changes in the service and leave aside changes in the market (La Notte et al., 2021). It does not state a deflator or a price basis for wood. The accounts therefore report both current and constant prices. Constant prices multiply the value of each year by the deflator of Table 12, which is the HICP of the EU27 (Eurostat, 2026b) rebased to 2010. A single EU27 index and the base year 2010 are a method choice that keeps all the accounts of this series on one price basis, and the INCA tool uses national indices with base year 2000 instead.

Table 12: Deflator applied to current prices, with 2010 equal to 1.

Year	Deflator
2000	1.272
2006	1.090
2012	0.947
2018	0.898
2024	0.717

Overuse and the sustainable flow take the same unit value as the NAI of their country and year, since the method has no separate price for them.

2.5 The spatial key

Part II spreads national flows over the forest of Corine Land Cover in proportion to Dry Matter Productivity (Vallecillo et al., 2019), and the Earth observation roadmap of the INCA project keeps a forest mask and that proxy (Zurbarán Nucci et al., 2025). The weight of a 100 m cell in year t is therefore

$$w_{i,t} = F_{i,t} \cdot P_{i,t} \tag{5}$$

where $F_{i,t}$ is 1 for forest in the Corine accounting layer of the year (European Environment Agency, 2026) and 0 elsewhere, and $P_{i,t}$ is the annual productivity resampled to 100 m with bilinear interpolation (see 05_prepare_weights.R in Annex I). A cell then receives

$$\text{Value}_{i,t} = \text{Value}_{c,t} \cdot \frac{w_{i,t}}{\sum_{j \in c} w_{j,t}} \tag{6}$$

so the cells of a country add up to its national value (see 10_write_maps_100m.R in Annex I).

Forest mask. Part II limits the mask to the broad-leaved, coniferous and mixed forest classes (311, 312 and 313) and leaves out transitional woodland and shrub (324) (Vallecillo et al., 2019). The INCA tool includes that class, so the two differ, and a decision was made to follow the report. Table 13 shows that the excluded class equals 13 percent of the mask in 2018.

Table 13: Area of the forest mask and of the excluded class in the EU27, in million hectares.

Year	Forest mask (311, 312, 313)	Transitional woodland and shrub (324)
2000	139.5	18.3
2006	138.5	19.3
2012	139.9	17.9
2018	139.7	17.9
2024	137.1	20.4

Availability. The mask holds all forest, available or not. Part II tried to delineate available forest through slope and protected areas and rejected this, because thresholds differ across countries and the result can mislead (Vallecillo et al., 2019). A map of available forest now exists for 2020 (European Commission, Joint Research Centre, 2026), and a decision was made to leave it out. It covers one year in a series of five, the reports do not call for it, and the statistic that is spread already refers to available forest.

Productivity. Part II and the roadmap name Dry Matter Productivity. This work uses Gross Dry Matter Productivity, a separate Copernicus product that also counts the autotrophic respiration of the plants, because it was the product at hand and the other needs a further bulk download.[1] The proxy only shares a national value among the cells of a country, so what matters is the relative pattern within a country and national totals do not depend on it. The 1 km product, version 2, serves 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-c), and the 300 m product, version 2, serves 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-d).

A test then measured what this choice costs (see 12_compare_dmp_gdmp.R in Annex I). It compared the Dry Matter Productivity products (Copernicus Land Monitoring Service, n.d.-a, n.d.-b) with half of the gross products, since the quality page of the 1 km product describes a constant fraction of 0.5. It covers 10 of the 180 pairs of files behind the key, namely the files ending on 10 January and 10 July of each reporting year, over the whole of Europe. The tolerance was set before the comparison. In every pair, the absolute difference could reach 1 percent of the productivity total, and the cells with data in only one product could reach 0.1 percent of the cells. Table 14 shows the outcome, which is that Dry Matter Productivity equals exactly half of the gross value in every cell, with data in the same cells. A constant ratio leaves the share of every cell unchanged, so on this evidence the two products give the same key, and the key stays on Gross Dry Matter Productivity. The other 170 pairs were not compared, so the finding rests on this sample.

[1]The annual layers are the sum of the 36 ten-day values of each year, each multiplied by its number of days (see 00_prepare_gdmp_annual.R in Annex I).

Table 14: Dry Matter Productivity against half of Gross Dry Matter Productivity, by pair of files over Europe. Cells are in millions, and the two gaps are percentages.

File date	Product	Cells with data	Data in one product only (%)	Absolute difference (%)	Within tolerance
2000-01-10	1km	13.7	0.0	0.0	Yes
2000-07-10	1km	19.1	0.0	0.0	Yes
2006-01-10	1km	13.2	0.0	0.0	Yes
2006-07-10	1km	19.2	0.0	0.0	Yes
2012-01-10	1km	13.7	0.0	0.0	Yes
2012-07-10	1km	19.1	0.0	0.0	Yes
2018-01-10	300m	134.4	0.0	0.0	Yes
2018-07-10	300m	173.5	0.0	0.0	Yes
2024-01-10	300m	114.5	0.0	0.0	Yes
2024-07-10	300m	173.1	0.0	0.0	Yes

2.6 Regional tables

The value of a region is the national value times the share of the region in the weight of its country in that year, so the regions add up to the country and follow the same key as the maps (see 06_prepare_nuts_weights.R in Annex I for the shares and helpers_results.R in Annex I for the regional values). Shares are computed on the 1 km weights, which keeps the step in the repository. All regions of a country therefore share one unit value, and the overuse of a country reaches all its regions in proportion to their weight, since the data do not show where removals take place inside a country.

2.7 Maps

The maps use the projection EPSG:3035 on a 100 m grid aligned with the Corine grid. There are six per year, for the actual flow, the sustainable flow and the overuse, in physical terms and in constant prices, which gives 30 maps. They hold whole numbers per cell, in cubic decimetres and in euro cents, which makes the files several times smaller than decimals with no loss that matters at this resolution. Because they are too large for the repository, they are kept in an external folder with a README file. The 1 km maps of the flow and its value are kept outside the repository as well, and they give the figures of this report (see 09_write_maps_1km.R in Annex I).

2.8 Checks

Each script stops when a check fails and records the result. Table 15 lists them with the largest relative gap found. The 100 m maps are read back from disk and compared with the national values country by country (see 11_check_maps_100m.R in Annex I). The workbook also carries a sheet that compares the NUTS2 and NUTS3 totals with the NUTS0 totals for every table, metric and year.

Table 15: Checks in the scripts. A script stops if a gap exceeds its tolerance.

Check	Largest gap	Tolerance
Sustainable flow and overuse add up to removals	1.5e-16	1e-06
One row per member state and year	0.0e+00	1e-06
Constant prices return to current prices through the deflator	2.0e-16	1e-06
The deflator equals one in the base year	0.0e+00	1e-06
1 km maps of the flow against the national tables	2.9e-11	1e-04
1 km maps of the value against the national tables	4.7e-11	1e-04
100 m physical maps against the national tables, by country	6.8e-05	1e-03
100 m monetary maps against the national tables, by country	4.9e-05	1e-03

3 Results

3.1 Actual flow

Table 16 shows the accounts for the EU27. The NAI falls by 22 percent between 2000 and 2024. Eurostat's own aggregate for available forest falls too, from 891 million cubic metres in 2000 to 784 million in 2018, by 12 percent, so part of the fall is present in the statistics themselves, and the changes of source described in Section 2.2.2 add to it.

Table 16: Physical accounts of the EU27, in million cubic metres. The last column counts Member States with overuse.

Year	Actual flow (NAI)	Removals	Sustainable flow	Overuse	Countries with overuse
2000	955.3	464.3	464.3	0.0	0
2006	915.6	476.2	476.2	0.0	0
2012	887.5	464.0	464.0	0.0	0
2018	822.2	562.1	562.1	0.0	0
2024	749.1	525.0	518.8	6.2	3

Figure 1 shows where the flow is placed. Borders between countries can appear on the map, because each country spreads its own national total.

3.2 Sustainable flow and overuse

Removals grow from 464 million cubic metres in 2000 to 562 million in 2018, and they take 68 percent of the NAI in that year against 49 percent in 2000. Figure 2 shows the ratio by Member State. Table 17 lists the cases of overuse in the reporting years. All of them fall in 2024, and both sides of each comparison repeat earlier years, so they describe the last reports of those Member States and may not describe 2024 itself. Czechia sits almost exactly on the threshold, with a ratio of 1.002. Looking at every year of the series, overuse appears in 12 country-years between 2019 and 2023, so the reporting years understate how often the threshold has been crossed recently.

Table 17: Member States with overuse in the reporting years, in thousand cubic metres.

Year	Country	NAI	Removals	Overuse	Removals / NAI	NAI origin	Removals origin
2024	EE	8,200.0	11,564.0	3,364.0	1.410	carried	carried
2024	LV	16,126.6	18,920.0	2,793.4	1.173	carried	carried
2024	CZ	20,458.3	20,495.7	37.4	1.002	carried	carried

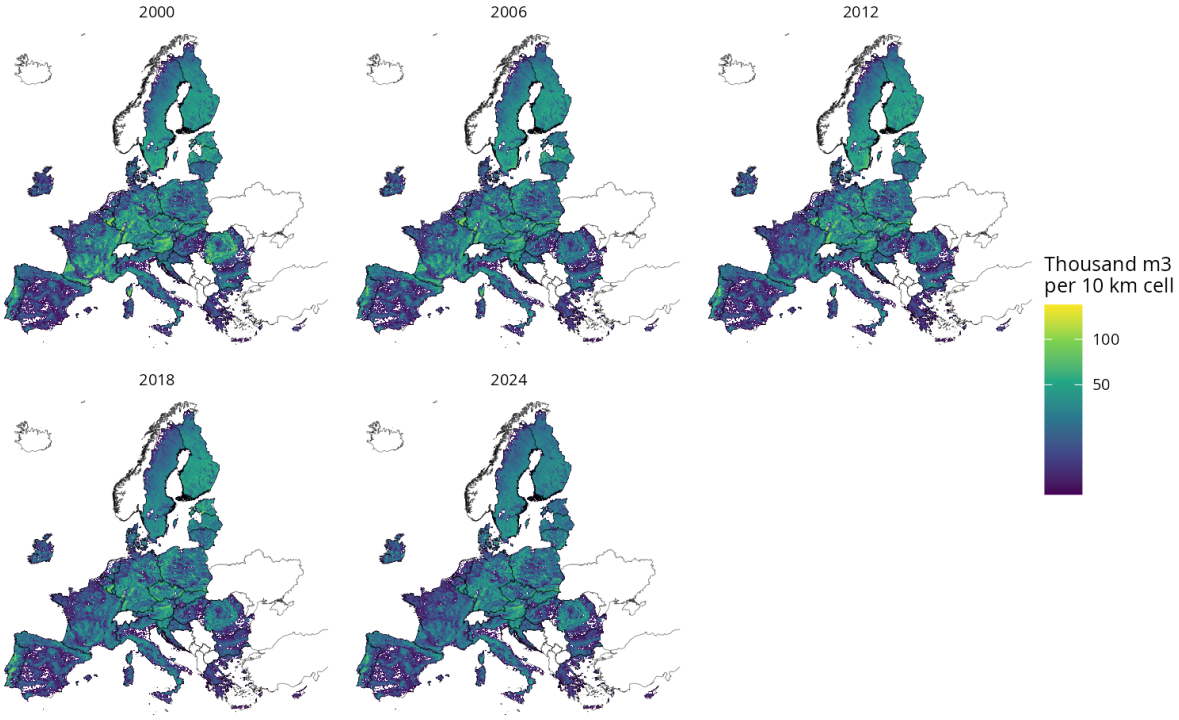


Figure 1: Actual flow of wood provision, in thousand cubic metres per 10 km cell, shown as the sum of the 1 km cells.

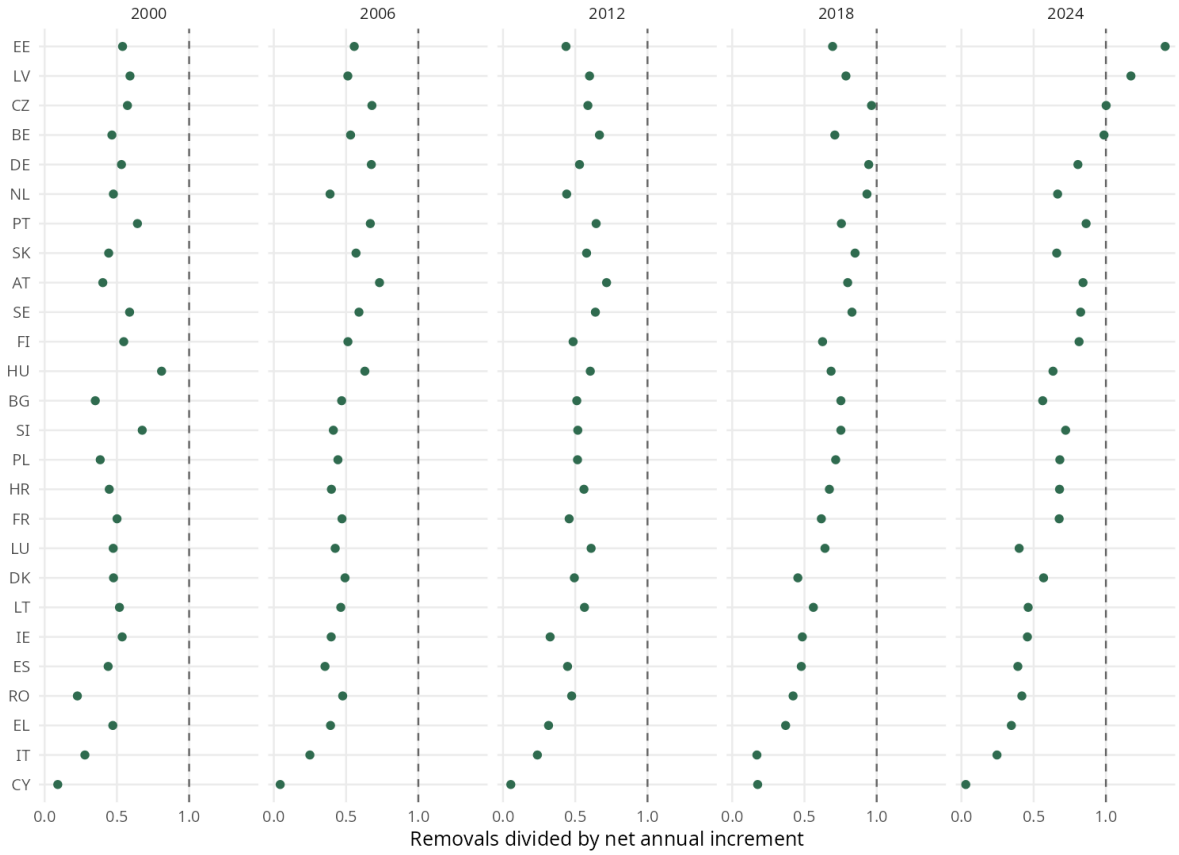


Figure 2: Removals divided by the net annual increment, by Member State. Points to the right of the dashed line are overuse. Malta has no values.

3.3 Value

Table 18 gives the value of each quantity. In current prices the flow rises by 19 percent over the period, and in constant prices it falls by 33 percent, since the deflator removes the general rise in prices and the NAI itself falls. Figure 3 maps the flow in constant prices. Each cell of a country has the same unit value, so within a country this map has the pattern of the physical map. Figure 4 shows the unit values, whose median in 2018 is 24.3 euro per cubic metre.

Table 18: Monetary accounts of the EU27, in million euro.

Year	Actual flow, current prices	Actual flow, 2010 prices	Sustainable flow, 2010 prices	Overuse, 2010 prices
2000	20,142	25,623	12,188	0
2006	18,680	20,354	10,058	0
2012	22,319	21,146	11,119	0
2018	21,213	19,054	13,269	0
2024	23,929	17,167	12,224	146

3.4 Comparison with the published accounts

Table 19 compares the value in current prices with the annexes of INCA Part III, for the same 27 Member States. The annexes do not state their price basis. Their values for 2012 match our current prices closely, which would also fit a constant series based on 2012, so the comparison of earlier years is the less certain one. The year 2012 lies within 0.3 percent of the published total, and 2000 and 2006 each lie about 12 percent below it. Several large countries agree closely in 2012, as Germany at 3,626 million euro against 3,626 million and Sweden at 3,581 million against 3,520 million. The largest relative gap in that year is Croatia, at 219 million euro against 40 million, which comes from the export point of 2012 removed above. Part III holds an earlier release of the same statistics, so an exact match is neither possible nor expected.

Table 19: Value of the actual flow in current prices against INCA Part III, EU27, in million euro.

Year	Accounts	INCA Part III	Median difference by country (%)
2000	20,142	22,871	11
2006	18,680	21,125	13

Year	Accounts	INCA Part III	Median difference by country (%)
2012	22,319	22,386	9

4 Limitations

Statistics. Values reported for available forest cover few countries before 2018, and most values before then are model estimates of total forest, so they probably overstate the flow. Changes of source move the series for reasons that have nothing to do with forests, as Table 7 shows, so comparisons over time within those countries need care. The carbon budget series ends in 2019 and is flat from 2016, and the values of 2024 mostly repeat earlier years. They should be refreshed when Eurostat holds more of that year.

Overuse. The removals of available forest are compared with a NAI that in some countries refers to total forest, which understates overuse in those countries. The ratio could cross the threshold in more countries if the model estimates were scaled to available forest. Regional overuse is the national overuse shared by the key, so it does not show where overharvesting occurs inside a country. A map of overuse has the same limit.

Value. Few countries report the timber value before 2012, so most early values come from an export price divided by a mean ratio, and the monetary series is better read as an estimate than as a measurement. 10 Member States take the EU mean, of which 7 lost their timber values in the screening, so they all share one ratio. A country with low export volumes can have an unrepresentative export price. Every cell and region of a country has one unit value, so the maps do not show price differences within a country.

Space. The key combines forest with the productivity of the vegetation, and productivity measures growth of vegetation in general, so some flow is placed on forest that is not harvested. Productivity comes from a 1 km product in 2000, 2006 and 2012 and from a 300 m product in 2018 and 2024, so spatial detail differs between the two groups and changes in the pattern between them partly reflect the change of product. The equivalence of the gross and net products rests on a sample of 10 of the 180 pairs of files, as the Methods section explains. The Corine accounting layers have a minimum mapping unit of 25 hectares. Regional shares use weights at 1 km, whose cells on borders are assigned by their centre, so they can differ slightly from a sum of the 100 m maps.

Appendix: accounts by Member State

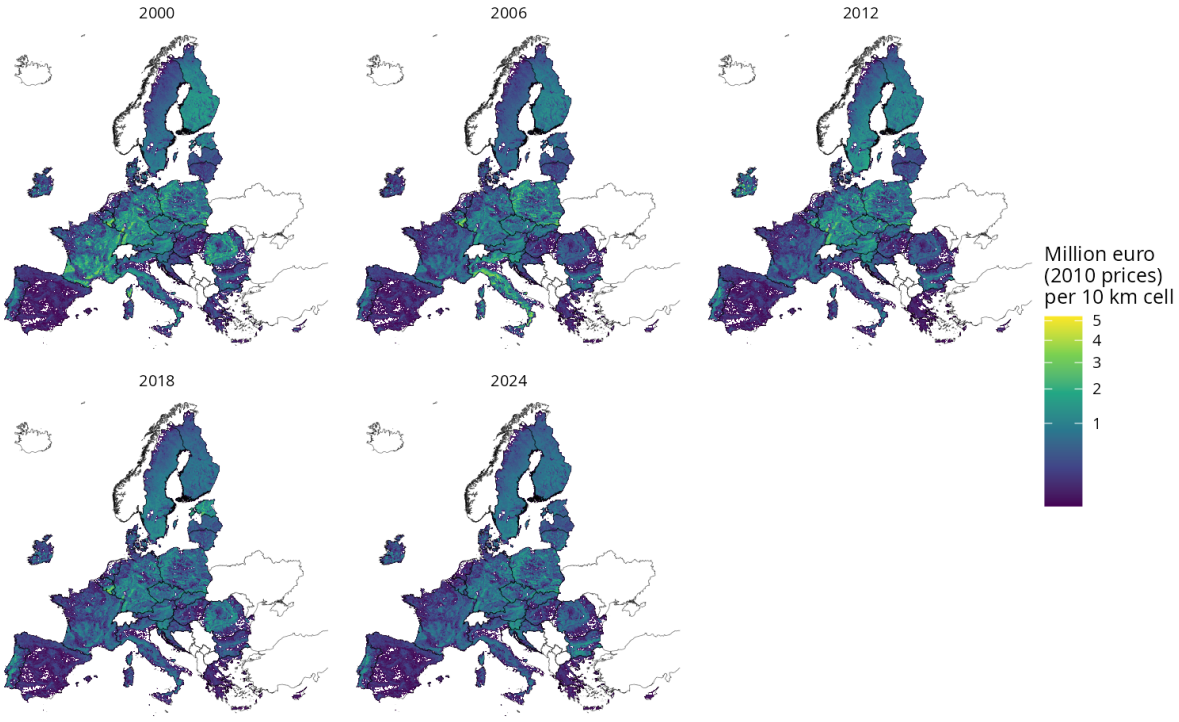


Figure 3: Value of the actual flow, in million euro at 2010 prices per 10 km cell.

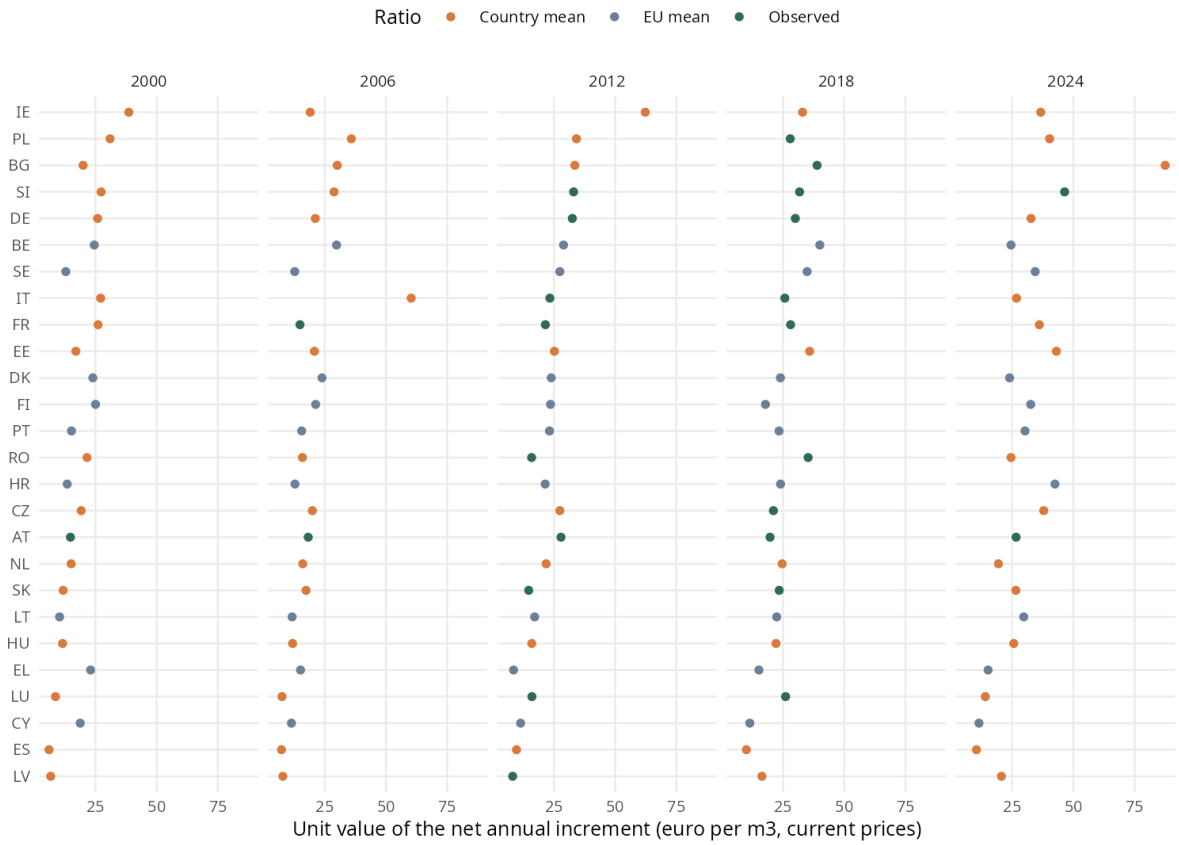


Figure 4: Unit value of the net annual increment by Member State, in current prices, with the origin of the ratio.

Table 20: Actual flow and its value by Member State in 2000, 2012 and 2024. NAI is in thousand cubic metres and value in million euro at 2010 prices.

Country	NAI 2000	NAI 2012	NAI 2024	Value 2000	Value 2012	Value 2024
AT	36,936	29,497	27,960	695	780	533
BE	10,880	8,630	5,300	339	237	93
BG	15,683	15,454	9,680	398	491	607
CY	280	254	55	7	3	0
CZ	28,597	32,106	20,458	697	835	556
DE	113,545	111,379	90,144	3,737	3,435	2,112
DK	6,925	7,747	9,783	211	176	168
EE	18,598	18,583	8,200	402	444	253
EL	5,938	3,984	4,905	174	32	53
ES	39,404	43,679	46,457	302	404	347
FI	114,040	123,474	89,264	3,630	2,766	2,085
FR	153,973	93,230	73,914	5,105	1,904	1,913
HR	8,204	10,202	8,508	141	207	259
HU	8,933	12,042	11,204	131	183	206
IE	5,498	8,703	9,343	270	514	246
IT	41,286	38,182	41,427	1,425	846	795
LT	11,998	13,857	17,292	158	225	368
LU	622	760	819	7	12	8
LV	27,335	23,133	16,127	234	180	238
MT	0	0	0	0	0	0
NL	2,565	2,595	1,387	49	54	19
PL	75,029	79,820	76,143	2,955	2,589	2,200
PT	20,965	20,660	17,467	406	454	379
RO	63,163	35,818	40,488	1,731	540	711
SE	125,520	130,469	102,990	2,057	3,393	2,540

Country	NAI 2000	NAI 2012	NAI 2024	Value 2000	Value 2012	Value 2024
SI	3,649	6,950	7,039	127	218	234
SK	15,732	16,252	12,759	236	227	242

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Annex I. Scripts of the pipeline

This annex describes each script in the order in which it runs. The link under each name opens its code on GitHub, and the paragraph explains what the script takes in, what it does and what it saves for the steps that follow. Script 00 builds the annual productivity layers and runs only by hand. Scripts 01 to 06 prepare the inputs. Script 02 needs internet access, and the others need source files that are kept outside the repository. Scripts 07 to 09 run from the repository alone. Scripts 10 and 11 write and check the 100 m maps, and they need large files kept outside the repository again. Script 12 is a test that runs by hand, and the report reads the small table it saved.

00_prepare_gdmp_annual.R

View on GitHub

This script builds the annual productivity layers from the ten-day files of Gross Dry Matter Productivity, using the 1 km product for 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-c) and the 300 m product for 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-d). It reads the part of each file that covers Europe from the Copernicus Data Space Ecosystem, multiplies the daily rate by the days that the file covers, counts missing cells as zero and adds up the 36 files of the year. It saves one file per year, `gdmp_<product>_<year>_4326.tif`, outside the repository for script 05. The pipeline does not run this script every time, because it needs Copernicus access keys, reads 180 files over the network for hours and uses many gigabytes of temporary disk space, while its five results take about 1 GB and rarely need replacing. Its line is therefore commented out in run_all.R, and it skips any layer that already exists.

01_prepare_nuts.R

View on GitHub

This script takes the NUTS 2024 boundaries of Eurostat at a scale of 1:1 million (Eurostat GISCO, 2024), in the copy distributed with the INCA plugin (European Union, 2026), and keeps only the level of countries. It marks the 27 Member States and stops if it does not find all of them. The result is a small file, `NUTS_RG_01M_2024_3035_LEVL0.gpkg`, which the later scripts use to identify the countries and to draw the maps, and the copyright notice of the boundaries is copied next to it.

02_download_eurostat.R

View on GitHub

This script downloads four tables through the Eurostat data service. From `for_vol_efa` (Eurostat, 2026e) it keeps the net annual increment and the removals of every forest class, in thousand cubic metres. From `for_sup_cp` (Eurostat, 2026d) it keeps the output of forest trees, in million euro. From `for_basic` (Eurostat, 2026c) it keeps the exports of roundwood of all species, in value and in volume. The first three are limited to 2000 to 2024. The fourth table is the annual price index of the EU27 for all items, with 2015 as its base (Eurostat, 2026b). Each table is saved as an rds file, namely `for_vol_efa_thsm3.rds` for the physical accounts and `for_sup_cp_p1tr.rds`, `for_basic_rw_exports.rds` and `hicp_eu27.rds` for the monetary accounts. A new release of these tables changes the results, so this step runs only when an update is intended.

03_prepare_carbon_budget.R

View on GitHub

This script reads the workbook that combines the carbon budget model with the forest accounts (Pilli & Grassi, 2021), which it downloads if the file is missing. The workbook has one sheet per country and none for Malta. For each of the 26 other Member States and each year from 2000 to 2019, the script keeps the total net annual increment and the share of available forest in the growing stock. It stops if a value is missing or if an increment is not positive. The result is `cbm_nai.rds`, a small table that fills the gaps of the physical accounts in script 07.

04_extract_manual_annex.R

View on GitHub

This script reads the text of INCA Part III (La Notte et al., 2021) and extracts the timber tables of Annexes AA37 to AA47, which give the supply by Member State in 2000, 2006 and 2012, in million cubic metres and in million euro. The annexes include the United Kingdom, so the script stops unless it finds 28 countries in each of the three years. The result is `inca_part3_timber_annex.csv`, short enough to check by eye against the published annexes. Scripts 07 and 08 use it only for comparison, so no result of the accounts depends on it.

05_prepare_weights.R

View on GitHub

This script builds the spatial key for each reporting year. It takes the Corine Land Cover accounting layer at 100 m (European Environment Agency, 2026) and the annual Gross Dry

Matter Productivity, from the 1 km product in 2000, 2006 and 2012 (Copernicus Land Monitoring Service, n.d.-c) and from the 300 m product in 2018 and 2024 (Copernicus Land Monitoring Service, n.d.-d). Script 00 builds these annual layers, and `manifest_heavy_inputs.csv` lists them with their sources. The script marks the forest classes 311, 312 and 313, resamples the productivity to the 100 m grid and multiplies the two, so cells outside forest or outside the EU27 receive zero. The 100 m weights take about 2.4 GB for the five years, so they stay outside the repository for script 10. The script adds them up in cells of 1 km and saves each year as `clc_gdmp_weight_1km_<year>.tif`, a file of about 11 MB that scripts 06 and 09 read. It also saves the hectares of forest and of the excluded class 324 as `forest_mask.rds` for this report.

06_prepare_nuts_weights.R

View on GitHub

This script takes the boundaries of the countries, NUTS2 and NUTS3 regions of the EU27 (Eurostat GISCO, 2024), in the same copy as script 01, and the 1 km weights of script 05. For each region and reporting year, it adds up the weight of the cells that fall inside the region and divides it by the total of all regions of that level in the country. The result is `weight_by_nuts.rds`, which script 08 uses to share national values among the regions. The number of regions at each level is saved as `region_counts.rds` for this report.

07_build_physical_accounts.R

View on GitHub

This script takes the forest volumes of script 02, the carbon budget estimates of script 03 and the published annexes of script 04. It builds the net annual increment of each Member State and year from 2000 to 2024 from the value reported for available forest, then the carbon budget estimate, then the mean of the nearest values of the country, and at the ends of the series the nearest value. Removals follow the same order, with reported removals of total forest in place of the estimate. Malta takes zero. The script then compares removals with the increment to obtain the overuse and the sustainable flow, and it stops if a value is missing or if the two quantities do not add up to the removals. It also tests the candidate rules for the gaps against the annexes. It saves `wood_nai_faws_country_year.rds` and `wood_overuse_country_year.rds` for the later scripts, and it saves the small tables of the physical results in `data/outputs/report/`, with the figure of the ratio of removals to increment.

08_build_monetary_accounts.R

View on GitHub

This script takes the physical accounts of script 07, the output of forest trees, roundwood exports and price index of script 02, the regional shares of script 06 and the annexes of script 04. It computes the export price of each country and year and drops the points more than three times away from the median of the country. It removes the outliers of the INCA tool (European Union, 2026) and the countries and points whose ratio falls outside the range of Section 2.4.1, and it fills the gaps with country and EU means. The export price divided by the ratio gives the unit value, and the unit value times the physical quantity gives the value, in current prices and in constant 2010 prices. The script stops if a value is missing or if the deflator does not return to current prices. It saves `wood_value_country_year.rds` for the maps. It also writes the workbook of the Downloads page, `wood_provision_results.xlsx`, with four tables at three NUTS levels, the origin of each national value and a sheet that confirms that regions add up to countries. Finally, it saves the small tables of the monetary results and the figure of the unit values.

09_write_maps_1km.R

View on GitHub

This script takes the 1 km weights of script 05 and the national tables of scripts 07 and 08. For each reporting year, it gives each cell the national value times its weight divided by the total weight of the country, which yields 10 maps, five of the flow in thousand cubic metres and five of its value at 2010 prices. The maps take about 110 MB and stay outside the repository. The script reads them back and stops if the EU totals differ from the tables by more than 0.01 percent or if a country with a value has no forest weight. It saves the record as `maps_1km_totals.rds` and `checks_maps_1km.rds`, and it draws the two map figures of this report at 10 km.

10_write_maps_100m.R

View on GitHub

This script takes the 100 m weights of script 05 and the national tables of scripts 07 and 08. It spreads each national value in the same way as script 09 and writes 30 maps, which are the actual flow, the sustainable flow and the overuse in cubic decimetres and in euro cents at 2010 prices, for each of the five reporting years. It stores whole numbers per cell and picks up where it stopped if a run is interrupted. The maps take about 5.4 GB in total, so they stay outside the repository and are published in the external folder of the Downloads page.

11_check_maps_100m.R

View on GitHub

This script reads the 30 maps at 100 m back from disk, adds up their cells by country and compares the totals with the national tables. It stops if a country differs by more than 0.1 percent, and it leaves out of the comparison the countries whose national value in the map is under 10,000 of its units, where rounding to whole numbers has a large relative effect. Because the maps stay outside the repository, the script saves this record as `maps_100m_totals.rds` and `checks_maps_100m.rds`, and the report reads it from there.

12_compare_dmp_gdmp.R

View on GitHub

This script tests whether Dry Matter Productivity (Copernicus Land Monitoring Service, n.d.-a, n.d.-b) would change the spatial key. It reads the part that covers Europe of the net and gross files ending on 10 January and 10 July of each reporting year, from the Copernicus Data Space Ecosystem, and compares the net value with half of the gross value, cell by cell. It records how much the two differ and how many cells have data in only one product, against a tolerance set beforehand. The result is `dmp_gdmp_check.rds`, which this report reads. The script needs Copernicus access keys and stays out of run_all.R, and the comparison took about four minutes.

run_all.R

View on GitHub

This script runs the numbered scripts in order, grouped into four stages, which are the inputs (scripts 01 to 06), the tables (07 and 08), the 1 km maps (09) and the 100 m maps (10 and 11). With no stage named it runs the tables and the 1 km maps, which need only the repository. Script 00 appears as a commented line with a note that explains why it stays out of the run.

Helper files

View on GitHub: helpers_common.R, helpers_wood.R, helpers_monetary.R, helpers_weights.R, helpers_results.R and helpers_report.R

These files hold the steps that several scripts share, and they do not run on their own. The first holds the reporting years, the list of Member States, the saving of small tables and figures, the checks and the national values of the 100 m maps. The second fills the gaps in a series, compares removals with the increment and spreads national values over the 1 km grid. The third fills gaps with means and prepares the deflator. The fourth holds the forest classes, the access to the Copernicus store, the 100 m grid and the weight of each cell. The fifth shares national values among the regions and builds the workbook, and the sixth reads the small tables and formats them for this report.